

Phishing Email Detection and Awareness Report

Cybersecurity Task 1 – Future Interns

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Assessment Type: Phishing Email
detection Assessment

INTRODUCTION:

Phishing is one of the most widespread and financially damaging cyber threats targeting individuals and organizations today.

In these attacks, cybercriminals manipulate users into:

- Opening malicious links
- Downloading harmful attachments
- Revealing confidential information such as passwords or OTPs

Most phishing incidents occur not due to system weaknesses, but because users fail to recognize deceptive communication.

AIM OF THE TASK:

This task is designed to:

- Study real-world phishing email examples
- Detect warning signs of phishing attempts
- Categorize the risk level of suspicious emails
- Suggest corrective actions and preventive measures
- Build awareness about email-based cyber threats

Tools & Resources Used:

Email Analysis Tools:

- Sample phishing email datasets (public sources)
- Google Email Header Analyzer
<https://toolbox.googleapps.com/apps/messageheader/>
- MXToolbox Email Header Lookup
<https://mxtoolbox.com/EmailHeaders.aspx>
- Browser-based URL inspection techniques

Phishing Email Samples, Analysis & Response:

Sample 1: Fake Bank Alert Message

Full Email Content

From: security-alert@yourbank-secure.com

Subject: Immediate Action Required – Account Suspension Notice

Dear Customer,

We have detected unusual activity on your bank account. Due to security reasons, your account will be suspended within 24 hours unless immediate verification is completed.

Please verify your identity immediately by clicking the link below:

<http://yourbank-secure-login.com/verify>

Failure to complete verification will result in permanent account suspension.

Thank you,

Bank Security Team

Why This is Malicious:

- The sender domain is not an official bank domain
- Creates fear + urgency (24-hour suspension threat)
- Uses a fake verification link
- Generic greeting (“Dear Customer”) instead of real name
- The link is unrelated to official banking portals

Risk Level: HIGH (Phishing Attempt):

Why High Risk?

- Targets financial credentials (bank login details)
- Can lead to account takeover and financial theft
- Uses strong psychological manipulation (fear tactic)

Sample 2: Fraudulent PayPal Notification

Full Email Content

From: service@paypal-secure-support.com

Subject: Action Needed – Payment Issue Detected

Hello User,

We detected a problem with your recent PayPal transaction. Your account has been temporarily limited.

To restore access, please confirm your identity here:

<http://paypal-verification-center.com/login>

If you do not complete verification within 12 hours, your account will be permanently restricted.

PayPal Security Team

Why This is Malicious:

- Fake domain impersonating PayPal
- Mimics official branding to gain trust
- Creates urgency (“12 hours only”)
- Redirects to fake login page to steal credentials
- No personalization or transaction details

Risk Level: HIGH (Credential Phishing):

Why High Risk?

- Targets payment account login credentials
- Can lead to unauthorized transactions
- Attacker may gain access to stored cards and funds

Sample 3: Fake Job Offer Email:

Full Email Content

From: hr-recruitment@global-careers-offer.com

Subject: Congratulations! You Have Been Selected

Dear Candidate,

Congratulations! You have been selected for the position of Software Assistant in our company.

Please find your offer letter attached and confirm your acceptance within 48 hours.

Download your offer letter here:

Offer_Letter.zip

Once opened, follow the instructions to proceed with onboarding.

HR Department

Why This is Malicious:

- Unexpected job offer without application
- Suspicious .zip attachment (common malware carrier)

- Unknown HR sender domain
- Pressure to act quickly (48 hours)
- No interview or selection process mentioned

Risk Level: MEDIUM to HIGH:

Why Medium–High Risk?

- Attachment may contain malware or ransomware
- Can infect system when opened
- May steal files, passwords, or system access

Sample 4: Password Expiration Scam:

Full Email Content

From: it-support@account-security-reset.com

Subject: Your Password Will Expire Immediately

Dear User,

Your account password is expiring today due to security policy updates.

To continue using your account without interruption, please reset your password immediately:

<http://secure-password-reset-login.com>

If you do not update within 1 hour, your account will be locked permanently.

IT Security Team

Why This is Malicious

- Passwords normally do NOT expire this way urgently
- Fake reset link designed to steal login credentials
- Creates panic using “1 hour deadline”
- Fake IT security impersonation
- No official company domain used

Risk Level: HIGH (Credential Theft Attack):

Why High Risk?

- Direct attempt to steal login credentials

- Can lead to full account compromise
- Often used in corporate attacks (BEC/phishing chains)

Summary Table :

Sample	Attack Type	Risk Level	Target
Bank Alert	Credential Phishing	High	Banking data
PayPal Scam	Financial Phishing	High	Payment accounts
Job Offer	Malware Phishing	Medium–High	System infection
Password Reset	Credential Theft	High	Account access

Working of a Phishing Attack:

1. Attacker sends deceptive email to target
2. User interacts with malicious link or file
3. Fake website collects login credentials
4. Attacker gains unauthorized access

Security Best Practices:

- Verify sender identity before responding
- Avoid clicking unknown or shortened links
- Check website URL carefully before login
- Enable Multi-Factor Authentication (MFA)
- Report suspicious emails immediately
- Do not open untrusted attachments

Safe Practices vs Risky Behavior

✓ Safe Practices

- Confirm authenticity of email sources
- Hover over links before clicking
- Use updated antivirus protection

Unsafe Practices

- Sharing OTPs or passwords via email
- Trusting urgent or threatening messages
- Opening unknown file attachments

Email Header Analysis Section:

This section includes technical analysis of email headers to identify:

- Sender authenticity
- IP address tracing
- Domain verification
- Routing path inspection

Conclusion:

Phishing attacks rely heavily on social engineering rather than technical vulnerabilities. Strengthening user awareness, practicing safe email habits, and following security guidelines significantly reduces the risk of such attacks.